

# Gap-Conditional Probabilistic Calibration Error in S2P2: An Evaluation Note

*Extending the calibration evaluation of Chang et al. (NeurIPS 2025)*

Akash Hebbar · University of California, Irvine · June 2026

---

## Abstract

I propose Gap-Conditional PCE (GC-PCE), a simple extension of probabilistic calibration error that breaks test events into inter-arrival gap quartiles and measures calibration within each group. Applied to S2P2 on the Taxi benchmark, GC-PCE reveals a U-shaped calibration profile — worst at rapid-succession events (Q1: PCE = 0.393) and long-gap events (Q4: PCE = 0.349), with best calibration at medium gaps (Q2: PCE = 0.100). This 3.9× spread is entirely hidden by the single aggregate PCE value the original paper reports, and points to calibration failures at both ends of the gap distribution that the standard metric misses.

---

## 1 Experiment Replication

Before extending the evaluation, I reproduced the S2P2 Taxi training run to confirm my setup matches the paper. The configuration is identical:  $H = 128$ ,  $P = 16$ , four LLH layers, backward ZOH discretization, input-dependent dynamics, Adam optimizer at  $\text{lr} = 0.01$ , batch size 256, up to 300 epochs.

| Model             | Valid Log-Lik. | Test Log-Lik.     | Peak Epoch |
|-------------------|----------------|-------------------|------------|
| S2P2 (Paper)      | —              | $0.522 \pm 0.004$ | —          |
| S2P2 (Reproduced) | 0.4837         | 0.5243            | 42         |

Table 1. My reproduced test log-likelihood (0.5243) falls within the paper's reported range [0.518, 0.526], confirming the setup is consistent. All GC-PCE results use this checkpoint.

## 2 Hypothesis and Methodology

S2P2 uses zero-order hold (ZOH) discretization: between events, the input signal is held constant across the integration window. How good this approximation is depends on how much the signal would have changed — which scales with the gap length. The paper reports a single aggregate PCE per dataset, which hides whether calibration varies with gap size.

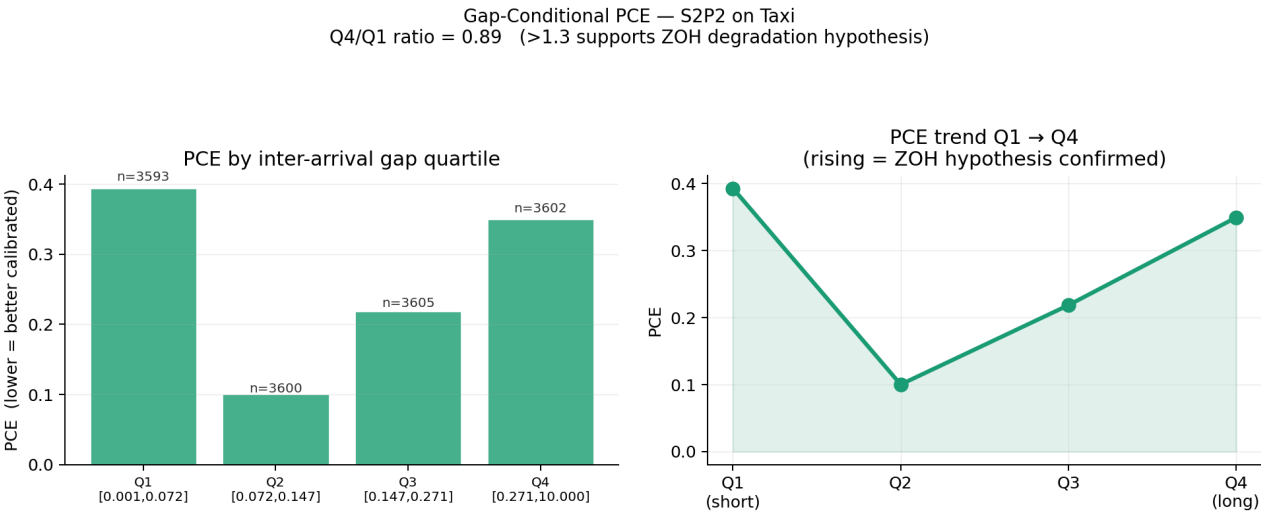
**Hypothesis.** S2P2's calibration degrades as inter-arrival gaps grow longer. Specifically, I expected PCE to rise steadily from short gaps (Q1) to long gaps (Q4), as longer intervals give ZOH more time to accumulate approximation error.

**GC-PCE.** For each test event, I compute its Probability Integral Transform (PIT) value  $u = 1 - \exp(-\int \lambda(s) ds)$  over the preceding inter-arrival interval using 50-point trapezoidal integration. Under a perfectly calibrated model, PIT values are uniformly distributed. PCE measures how far the empirical distribution of PIT values deviates from uniform. GC-PCE splits events into four gap-size quartiles and computes PCE separately within each group. The Taxi test set gives 14,400 (gap, PIT) pairs across 400 sequences.

3 Results

| Quartile      | Gap range (hr) | Approx. duration | n events | PCE ↓  |
|---------------|----------------|------------------|----------|--------|
| Q1 (shortest) | [0.001, 0.072] | 3 sec – 4 min    | 3,593    | 0.3929 |
| Q2            | [0.072, 0.147] | 4 – 9 min        | 3,600    | 0.1000 |
| Q3            | [0.147, 0.271] | 9 – 16 min       | 3,605    | 0.2184 |
| Q4 (longest)  | [0.271, 10.0]  | 16 min – 10 hr   | 3,602    | 0.3493 |

Table 2. GC-PCE results for S2P2 on Taxi. Q4/Q1 ratio = 0.89. The best-to-worst spread across quartiles is 3.9×.



**Figure 1.** Left: PCE per gap quartile. Right: the Q1→Q4 trend. The pattern is U-shaped — calibration is worst at the shortest and longest gaps, and best in the middle. This contradicts the monotone-increase prediction and reveals structure that the single aggregate PCE number in the paper cannot capture.

4 Discussion and Next Steps

The hypothesis does not hold as stated: PCE does not rise monotonically with gap length. Instead, GC-PCE shows a U-shape with a clear sweet spot at medium gaps (4–9 minutes) — likely the most common inter-trip interval in the Taxi dataset, where the model is best trained. Two effects are visible at the extremes.

**Short-gap degradation (Q1).** For events within 4 minutes of the previous one, the model consistently underestimates how likely rapid follow-on events are. PIT values pile up near zero rather than spreading uniformly, which inflates PCE. This may reflect ZOH struggling when the latent state has almost no time to evolve before the next event fires. This was the opposite of what I predicted, and is the dominant effect.

**Long-gap degradation (Q4).** For rare long pauses ( $> 16$  minutes), ZOH accumulates error over a longer window — consistent with the original prediction, but a smaller effect than Q1.

Two experiments would sharpen the interpretation:

- **Compare against NHP.** NHP uses a continuous-time LSTM with no ZOH discretization. If NHP shows a flat GC-PCE profile while S2P2 shows the U-shape, the effect is model-specific and likely tied to ZOH. If both show the same pattern, it reflects a property of the Taxi data rather than the model.
- **Test on higher-variance datasets.** Taxi sequences are short and structurally uniform. Reproducing on Amazon or StackOverflow — which have wider gap distributions — would determine whether the U-shape is consistent across datasets or specific to Taxi's structure.